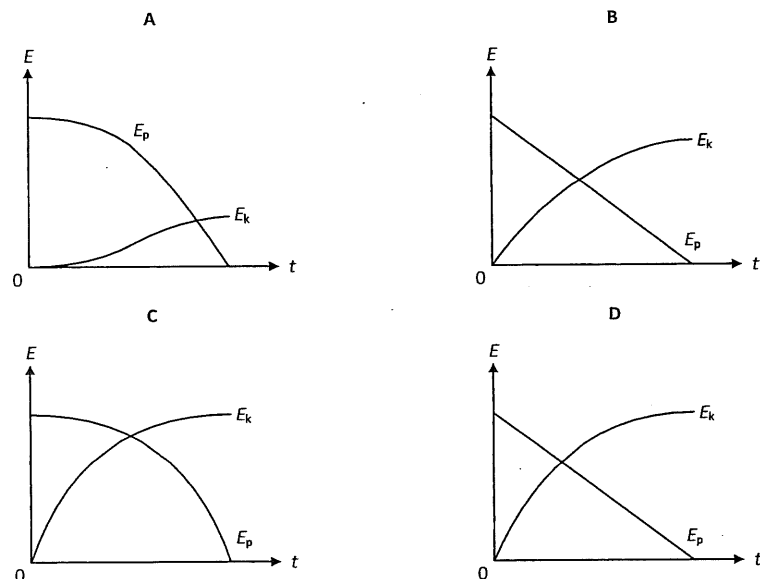


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/9]

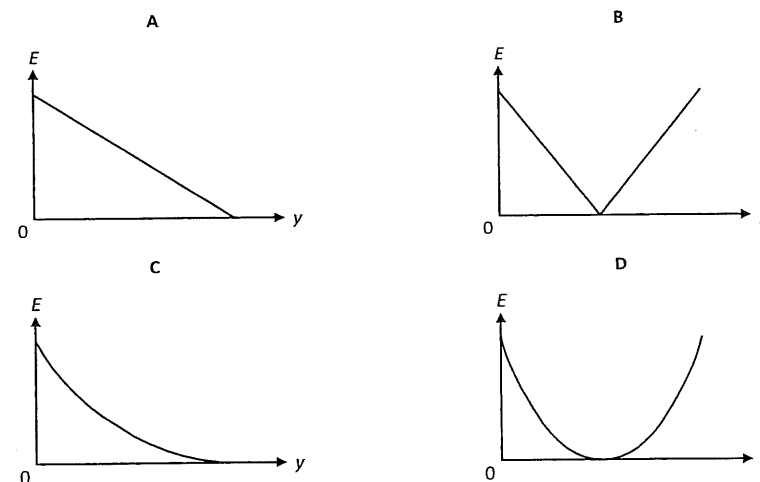
A steel ball is released at rest from the surface of oil in a tall cylinder so that it falls to the bottom of the cylinder. Which graph shows variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball?



2. [AJC/H2/Prelims 2018/P1/5]

A man throws a ball vertically upwards. The ball reaches a maximum height, and then falls back into the man's hand. Air resistance may be assumed to be negligible.

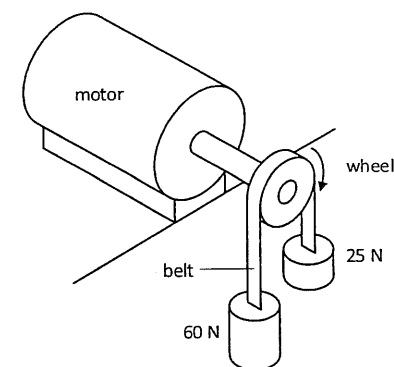
Which graph shows how the kinetic energy E of the ball varies with the vertical displacement y ?



3. [AJC/H2/Prelims 2018/P1/6]

The diagram shows an arrangement used to find the output power of an electric motor.

The wheel attached to the motor's axle has a diameter of 35 cm and the belt which passes over it is stationary when the weights have the values shown.

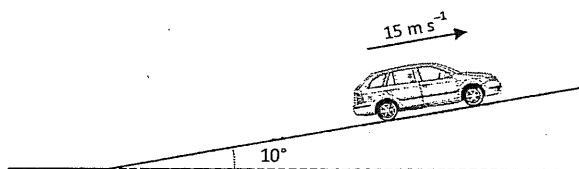


When the wheel is making 20 revolutions per second, what is the output power of the motor?

- A 250 W B 770 W C 1300 W D 1900 W

4. [DHS/H2/Prelims 2018/P1/7]

A car of mass 1100 kg is travelling at a constant speed of 15 m s^{-1} up a slope inclined at 10° to the horizontal. The combined frictional forces acting on the car are directed down the slope and are equal to $W/5$, where W is the weight of the car.



What is the useful output power of the car's engine?

- A 28 kW B 32 kW C 60 kW D 190 kW

5. [DHS/H2/Prelims 2018/P1/8]

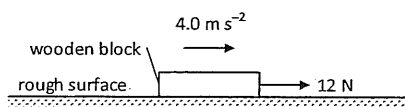
An old-fashioned 60 W lamp converts 95% of its energy supply into heat. A 4.0 W modern lamp has the same power output of light as the old-fashioned lamp.

What is the efficiency of the modern lamp?

- A 5.0% B 6.7% C 75% D 95%

6. [EJC/H2/Prelims 2018/P1/6 Modified]

A wooden block of mass 1.0 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at 4.0 m s^{-2} .



What percentage of work done is used to overcome frictional force when the block is displaced by 10 m?

- A 33.3% B 66.7% C 75.0% D 100%

7. [HCI/H2/Prelims 2018/P1/8]

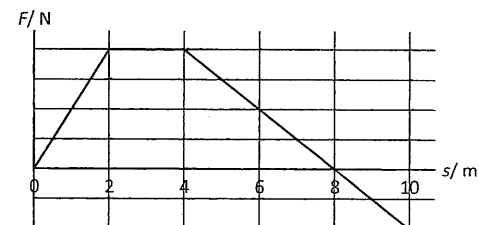
When a horizontal force F is applied to a trolley over a smooth horizontal surface for a distance x , its kinetic energy changes from 2 J to 6 J.

If a force $2F$ is applied to the trolley over a distance of $2x$, what is the final kinetic energy of the trolley if its original kinetic energy of the trolley is 2 J?

- A 12 J B 16 J C 18 J D 24 J

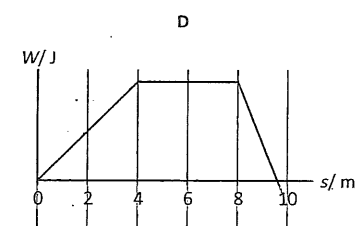
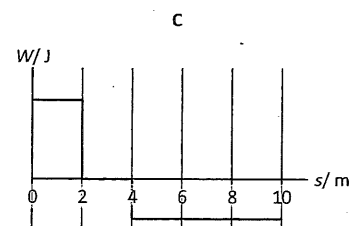
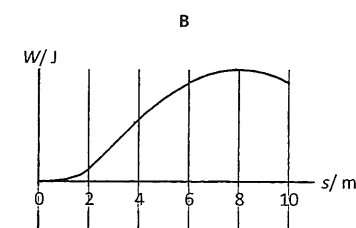
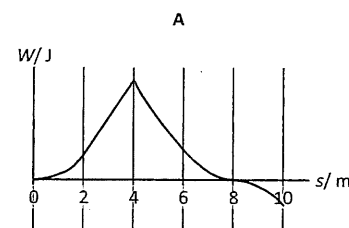
8. [HCI/H2/Prelims 2018/P1/9]

The figure below shows the variation the force F applied to an object with its displacement s .



Which of the following graphs correctly shows the variation of the work done W by force F on the object with respect to its displacement s ?

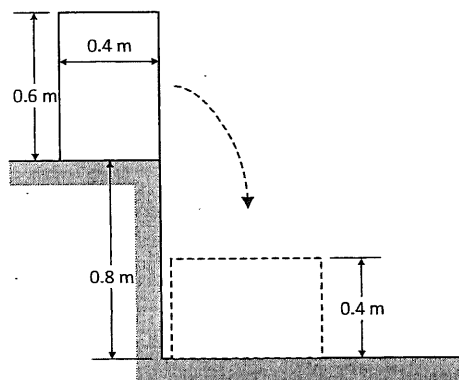
(Note the graphs in the options may not have the same scale for their vertical axes.)



9. [JJC/H2/Prelims 2018/P1/7 Modified]

A uniform solid block has weight 500 N, width 0.4 m and height 0.6 m. The block rests on the edge of a step of depth 0.8 m, as shown.

The block is knocked over the edge of the step and rotates through 90° before coming to rest with the 0.6 m edge horizontal.

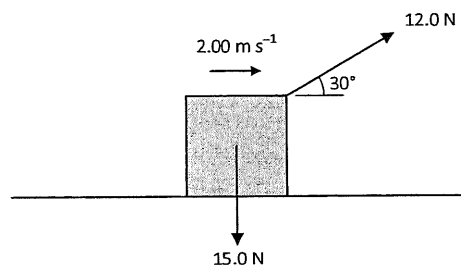


What is the decrease in gravitational potential energy of the block?

- A 300 J B 400 J C 450 J D 550 J

10. [JJC/H2/Prelims 2018/P1/8]

An object of weight 15.0 N is pulled along a horizontal surface at a constant velocity of 2.00 m s^{-1} . The force pulling the object is 12.0 N at 30° to the horizontal, as shown.

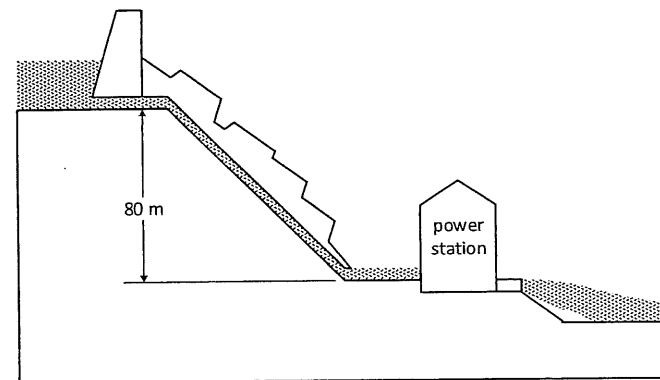


What is the power used to move the object?

- A 12.0 W B 20.8 W C 24.0 W D 30.0 W

11. [JJC/H2/Prelims 2018/P1/6]

A hydroelectric power station is shown below. Water is supplied from a reservoir which is 80 m above the power station. The water passes through its turbines at a volume-flow-rate of $6.0 \text{ m}^3 \text{ s}^{-1}$. The density of water is 1000 kg m^{-3} .

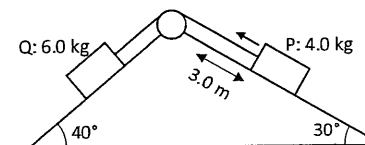


What is the electrical power output, if the efficiency of the power station is 60%?

- A 0.29 MW B 1.9 MW C 2.8 MW D 4.7 MW

12. [MI/H2/Prelims 2018/P1/10]

Two masses P and Q are connected by a light inextensible string which passes over a frictionless pulley. P and Q are initially at rest and their masses are 4.0 kg and 6.0 kg respectively. When Q is released, P moves up the plane inclined at 30° to the horizontal as shown.



What will be the speed of Q after P has moved 3.0 m up the plane?

- A 2.3 m s^{-1} B 3.3 m s^{-1} C 4.3 m s^{-1} D 5.3 m s^{-1}

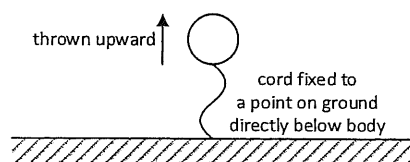
13. [MI/H2/Prelims 2018/P1/11]

A motorised boat moving with a constant speed S through still water experiences a total drag force X . What is the power of the motorised boat?

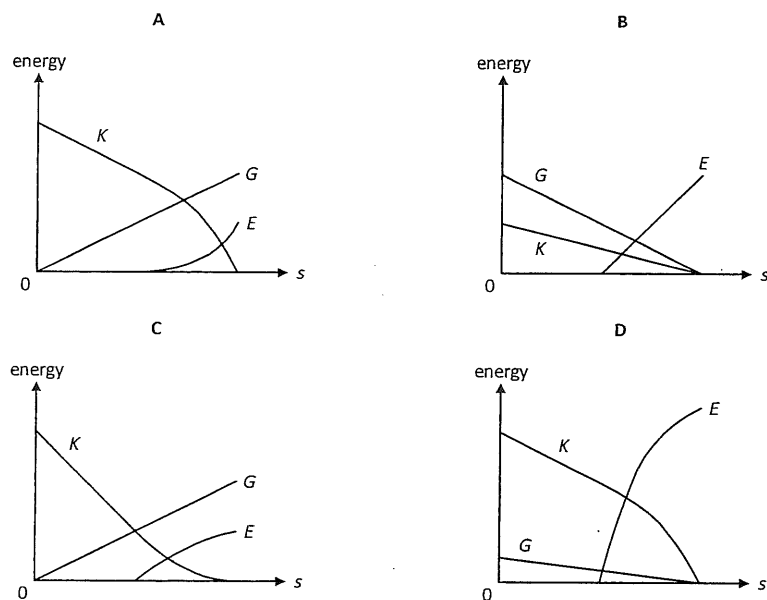
- A $X^2 S$ B $\frac{1}{2} X S^2$ C $X S$ D $\frac{1}{2} X S$

14. [MJC/H2/Prelims 2018/P1/6]

The diagram shows a body attached to an elastic cord being thrown vertically upwards. Initially the cord is unstretched but after a while it becomes stretched. The cord obeys Hooke's law and air resistance is ignored.

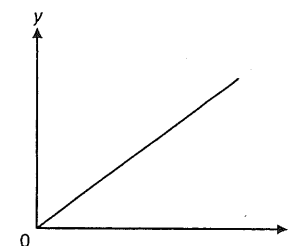


Which of the following shows the variation with displacement s of the kinetic energy K , gravitational potential energy G and elastic potential energy E ?



15. [NJC/H2/Prelims 2018/P1/6 Modified]

The graph shows the variation with a quantity x of a quantity y for a body that is falling in air at terminal velocity in a uniform gravitational field.

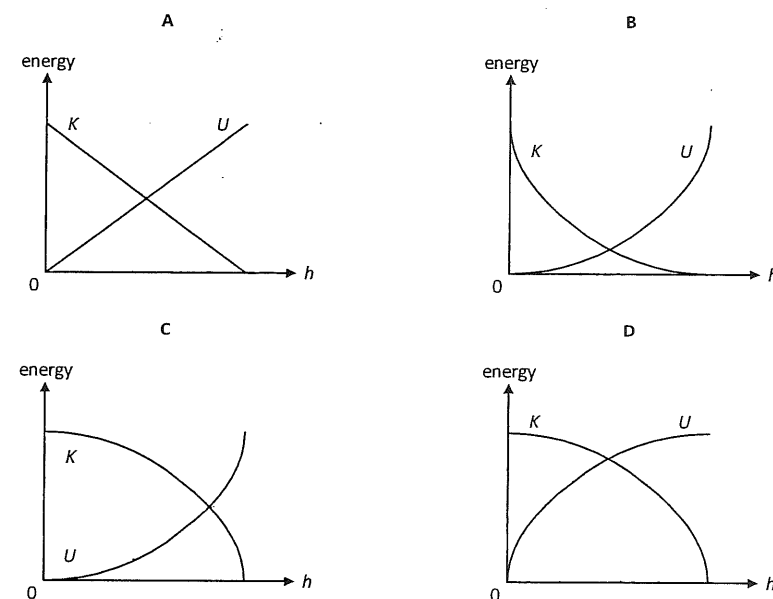


Which quantities could x and y represent?

	x	y
A	air resistance	acceleration
B	loss of height	gain in kinetic energy
C	loss of potential energy	work done against air resistance
D	time	velocity

16. [NYJC/H2/Prelims 2018/P1/8 Modified]

A ball is projected vertically upwards. At a height h , the kinetic energy of the ball is K and its potential energy is U . Taking air resistance to be negligible, which graph correctly shows the variations with h of K and U ?



17. [NYJC/H2/Prelims 2018/P1/9]

The engine of a car has maximum output power of 54 kW. The air resistance acting on the car when it is moving with speed v is cv^2 , where c is 2.0 kg m^{-1} .

What is the maximum speed that can be achieved by the car on level road?

- A 3.0 m s^{-1} B 5.0 m s^{-1} C 30 m s^{-1} D 160 m s^{-1}

18. [PJC/H2/Prelims 2018/P1/5]

The forward thrust provided by the engine of a car moving horizontally with constant velocity of 12 m s^{-1} on a straight road is 500 N.

Which of the following statements is correct?

- A The net force on the car is 500 N.
 B The average power of the engine is 3.0 kW.
 C The rate of work done by the engine is 6.0 kW.
 D The power of the engine is zero as the car is moving at constant velocity.

19. [RVHS/H2/Prelims 2018/P1/7]

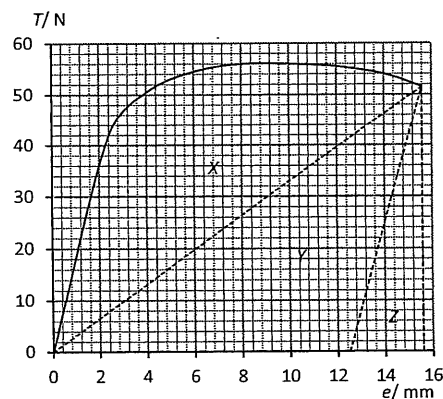
A mini rocket of mass 90 kg requires 28 kW of useful power in order to travel horizontally at a constant speed. The driving force on the rocket is given as 760 N.

What is the useful power required by the rocket to travel vertically at the same constant speed?

- A 28 kW B 33 kW C 61 kW D 68 kW

20. [SRJC/H2/Prelims 2018/P1/8]

The tension T in a sample of wire varies with extension e as shown in the diagram below.



The graph shows that the wire undergoes two types of deformation as it is extended to 15.6 mm.

In the region where Hooke's law applies, the deformation is elastic and the wire will lose this deformation when the tension is released.

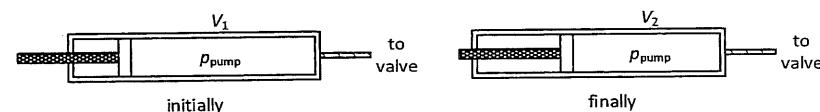
In the region where Hooke's law does not apply, the deformation is plastic and the wire retains this deformation when the tension is released.

Which area represents the elastic potential energy that is stored in the wire when it is extended to 15.6 mm?

- A Z
 B X + Y
 C Y + Z
 D X + Y + Z

21. [SRJC/H2/Prelims 2018/P1/9]

Air in a bicycle pump is forced through a valve at a constant pressure p_{pump} , to supply air to a tyre which is initially at a pressure p_{tyre} . In one stroke of the pump, the volume of air in the pump chamber is reduced from V_1 to V_2 .

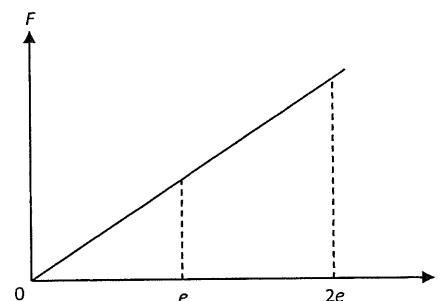


What is the work done on this air in one stroke of pump?

- A $p_{\text{pump}} V_1$ B $p_{\text{pump}} (V_1 - V_2)$ C $p_{\text{tyre}} V_1$ D $p_{\text{tyre}} (V_1 - V_2)$

22. [TPJC/H2/Prelims 2018/P1/11]

The extension x of a particular spring is related to the stretching force F as shown in the graph.



When the extension of the spring is e , the elastic potential energy stored in the spring is E .

What is the increase in the elastic potential energy when the extension is increased from e to $2e$?

- A E B $2E$ C $3E$ D $4E$

23. [TPJC/H2/Prelims 2018/P1/12]

An electric motor is required to haul a cage of mass 200 kg up a mine shaft through a vertical height of 800 m in 4.0 minutes.

What will be its electrical power required if its overall efficiency is 75%?

- A 0.89 kW B 4.9 kW C 5.2 kW D 8.7 kW

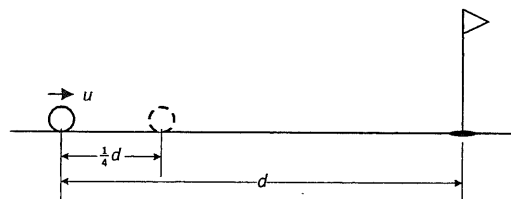
24. [VJC/H2/Prelims 2018/P1/6]

A truck of mass 3500 kg has an engine which can deliver a constant power. The truck experiences a constant frictional force of 3250 N. The truck travels at a velocity of 16 m s^{-1} when it is accelerating at 0.35 m s^{-2} . What is the maximum speed that the truck can achieve on level ground?

- A 6.0 m s^{-1} B 12 m s^{-1} C 20 m s^{-1} D 22 m s^{-1}

25. [YJC/H2/Prelims 2018/P1/6]

A golfer badly misjudges a hit, sending the ball only one-quarter of the distance to the hole. The hit gave the ball an initial speed of u , along a straight line on the grass.



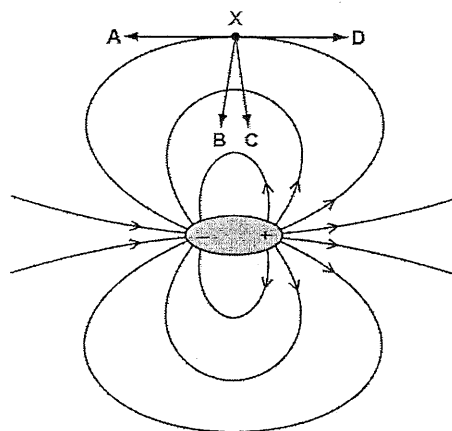
If the resistive force due to the grass is constant, what should be the minimum initial speed needed to get the ball to the hole from its original position?

- A $2u$ B $3u$ C $4u$ D $8u$

26. [AJC/H2/Prelims 2018/P1/20]

An electric dipole is made up of a pair of negative and positive charges of equal magnitude. The electric field of an electric dipole is shown below.

What is the direction of the force that acts on an electron placed at point X?



27. [VJC/H2/Prelims 2018/P1/5]

A 100 kg crate is pulled from rest across a floor with a constant force of 320 N. For the first 20.0 m, the floor is frictionless and for the next 10.0 m, a constant frictional force of 30.0 N acts on the crate.

What is the final speed of the crate?

- A 8.00 m s^{-1} B 8.37 m s^{-1} C 13.6 m s^{-1} D 13.9 m s^{-1}

II. Structured Questions (ACJC – YJC)

1. [CJC/H2/Prelims 2018/P2/1]

A hydroelectric power station could make a significant contribution to energy requirements.

Fig. 1.1 shows a dam at the hydroelectric power station which traps water behind a dam. When the height of the water behind the dam reaches 10.0 m, the water is released and passed through the turbines.

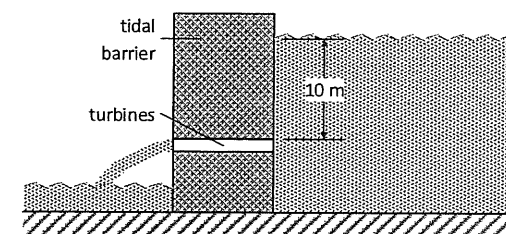


Fig. 1.1

- It takes 6.0 hours for a total mass of $1.32 \times 10^{12} \text{ kg}$ of water to flow through the turbines. The centre of mass of this amount of water is 5.0 m above the turbines. Calculate the loss in the potential energy of the trapped water when it is released through the turbines completely. Assume that the density of the water is uniform. [2]
- The potential energy calculated in part a. is lost and the efficiency of the power station is 40%. Calculate the average power output of the power station over this time period of 6.0 hours. [3]
- Suggest how the output power of the hydroelectric power station can be controlled as the level of trapped water decreases. [1]

[Total: 6]

2. [EJC/H2/Prelims 2018/P2/2]

Fig. 2.1 shows an archer with a bow.

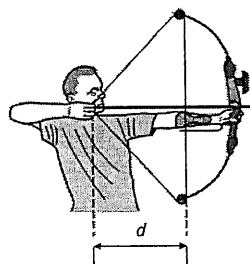


Fig. 2.1

The force F required to bend the bow and the corresponding displacement d of the arrow are measured. A plot of F against d is shown in Fig. 2.2.

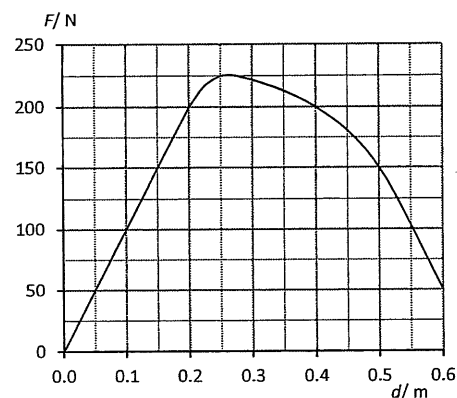


Fig. 2.2

- a. An experienced archer is able to draw an arrow further back, resulting in a greater displacement d of arrow as compared to a novice archer.

Using Fig. 2.2 and the principle of conservation of energy, suggest an advantage that this extra displacement provides. [2]

- b. i. Using Fig. 2.2, estimate the elastic potential energy stored in the bow when an arrow is displaced by 0.50 m. [2]

- ii. The arrow in b.i. is then released by the archer from the bow.

Determine the maximum possible speed of the arrow that has a mass of 3.5×10^{-3} kg. [2]

[Total: 6]

3. [EJC/H2/Prelims 2018/P2/7]

- a. For a steady and streamline flow of an incompressible and non-viscous fluid, the total hydraulic head H of the flow through a particular cross-section of a pipe is given by the expression

$$H = \frac{P}{\rho g} + \frac{v^2}{2g} + h,$$

where P is the pressure of the fluid at the cross-section, ρ is the density of the fluid, v is the velocity of the fluid at the cross-section, and h is the height of the centre-line of pipe above a reference level.

H is related the energy in the fluid. Suggest the type of energy associated with the following terms:

- i. $\frac{v^2}{2g}$, and [1]

- ii. h . [1]

- b. Fig. 3.1 shows a steady and fully developed flow through a horizontal pipe of varying diameter. The inlet has an inner pipe diameter D_A and fluid velocity v_A , while the outlet has an inner pipe diameter D_B and fluid velocity v_B .



Fig. 3.1

- i. Assuming there are no energy losses within the pipe, state how the total hydraulic head at the inlet and at the outlet are related. [1]

- ii. If the fluid is incompressible, show that v_A and v_B are related by the expression

$$\frac{v_A}{v_B} = \frac{D_B^2}{D_A^2}$$

[2]

- iii. The inner diameter of the inlet is 0.100 m and the inner diameter of the outlet is 0.050 m. Water flows into the inlet at a speed of 8.0 m s^{-1} .

Using the hydraulic head expression given in a. and your answers to b., determine the pressure difference between the inlet and outlet. (Density of water is 1000 kg m^{-3} .) [3]

- c. In a typical flow between 2 cross-sections of a pipe, there will be energy losses due to frictional forces between the walls of the pipe and the fluid. This is accounted for by including an additional head loss term,

$$H_{\text{loss}} = f_D \cdot \frac{v^2 L}{2gD},$$

where f_D is the friction factor, L is the length of the pipe, and D is the inner diameter of the pipe.

- i. State how this additional term will affect your answer to b.iii. [1]

- ii. If the pipe in b. has a length of 10 m and a friction factor of 0.0038, estimate the magnitude of the difference to your answer in b.iii. if frictional losses are considered. [2]

- d. A student suggests using water flow in rigid pipes to model blood flow in blood vessels.

- i. State a difference between water flow in rigid pipes and blood flow in blood vessels. [1]

- ii. The accumulation of fat and cholesterol deposits along the arterial walls has an effect of narrowing the arteries leading to an increase in blood pressure.

State and explain if the modelling proposed by the student will be able to correctly predict this observation. [3]

[Total: 15]

4. [JC/H2/Prelims 2018/P3/1]

A man travels on a toboggan down a slope covered with snow from point A to point B. The path is illustrated in Fig. 4.1.

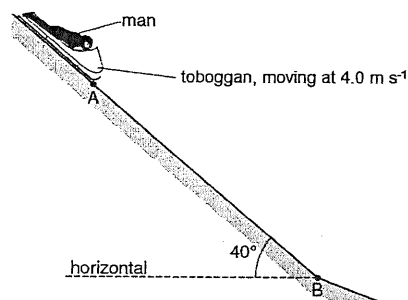


Fig. 4.1 (not to scale)

The slope AB makes an angle of 40° with the horizontal. Friction is not negligible.

The man and toboggan have a combined mass of 95 kg. They start with a speed of 4.0 m s^{-1} down the slope at A and have constant acceleration between A and B. They take 17.0 s to reach B with a speed of 37.0 m s^{-1} .

- Calculate the distance moved from A to B. [2]
 - The actual uncertainty in each measurement of the speed is 0.1 m s^{-1} . The percentage uncertainty in the time taken to move from A to B is 0.6%.
Calculate the percentage uncertainty in the distance moved from A to B. [2]
 - Use your answers in a.i. and a.ii. to determine the value of distance moved from A to B, with its actual uncertainty, to an appropriate number of significant figures. [1]
- For the man and toboggan moving from A to B,
 - calculate the change in kinetic energy, [2]
 - calculate the change in potential energy, [2]
 - hence determine the average frictional force that acts on the toboggan between A and B. [2]

[Total: 11]

5. [JC/H2/Prelims 2018/P3/2]

In a bungee jump, a light elastic rope is used to attach a man to a bridge. The man has a mass of 80.0 kg while the rope has a natural length of 25.0 m and an elastic constant of 120 N m^{-1} . The man steps off the bridge and falls vertically downwards from rest. Assume that air resistance acting on the man is negligible.

- Show that the extension of the rope is 6.54 m when the man is falling at maximum speed. [1]
 - Determine the maximum speed of the man after he steps off the bridge. [2]
 - Calculate the maximum kinetic energy of the man. [1]
- Calculate the maximum extension of the elastic rope when the man is at the lowest point of his motion. [2]
 - Hence, or otherwise, determine the gravitational potential energy of the man on the bridge. Assume that gravitational potential energy of the man is zero at the lowest point of the man's motion. [1]
- On Fig. 5.1, sketch three clearly-labelled graphs for the variation with downward displacement s of
 - the gravitational potential energy of the man (label as G),
 - the elastic potential energy stored in the rope (label as E), and
 - the kinetic energy of the man (label as K).

Assume that gravitational potential energy of the man is zero at the lowest point of the man's motion. Take $s=0 \text{ m}$ as the start point of motion.

[3]

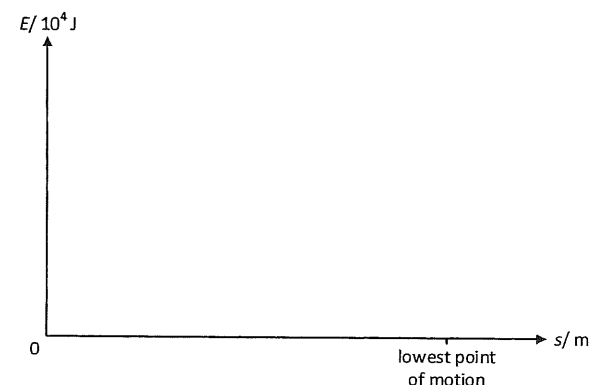


Fig. 5.1

[Total: 10]

6. [MI/H2/Prelims 2018/P2/2]

- Derive, from the equations for uniformly accelerated motion in a straight line, the expression for the kinetic energy E_k of a mass m moving at speed v ,

$$E_k = \frac{1}{2}mv^2$$

[3]

- A 2.0 kg box, lying on a frictionless incline that is at an angle of 40° above the horizontal, is connected by a cord that runs over a massless and frictionless pulley to a light spring of spring constant $k=120 \text{ N m}^{-1}$, as shown in Fig. 6.1. The box is released from rest along the incline when the spring is unstretched.

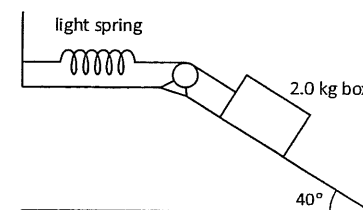


Fig. 6.1

- Calculate the energy stored in the spring when the box reaches 10 cm down along the incline. [1]
- Determine the distance D along the incline moved through by the box before it comes to a stop. [2]
- State what will happen to the answer calculated in (b)(ii) if the inclined angle is increased. [1]
- The frictionless incline is now replaced with a rough incline. Assuming that the friction between the rough incline and the box is 5.0 N, determine the new distance D' moved by the box before it comes to a stop. [2]

[Total: 9]

7. [NIC/H2/Prelims 2018/P3/1]

- a. Distinguish between *gravitational potential energy* and *elastic potential energy*. [1]
- b. A child's pogo stick comprises of a spring with spring constant of $1.50 \times 10^4 \text{ N m}^{-1}$. The combined mass of the child and pogo stick is 25.0 kg.

Fig. 7.1 shows the various instances of a child jumping vertically on the pogo stick.

At instance A, the spring compression is a maximum and the child is momentarily at rest. The footrest is 0.200 m below the reference level.

At instance B, the spring is relaxed and the child is moving upward.

At instance C, the child is again momentarily at rest and at the top of the jump.

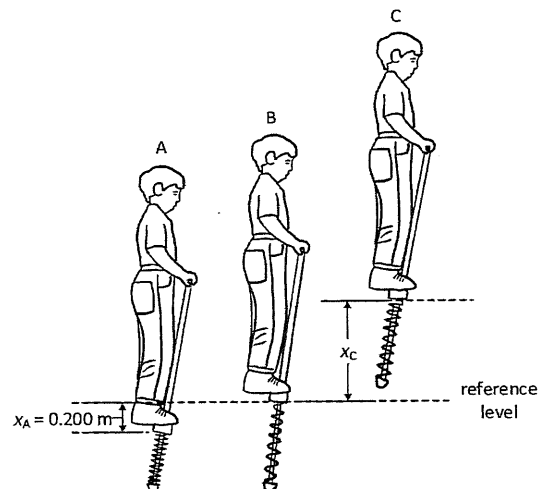


Fig. 7.1

- State the energy changes from instance A to C. Numerical values are not required. [3]
- Determine the value for x_C . [2]
- Determine the maximum speed of the child. [3]
- State and explain the position of the footrest, with respect to the reference level, when maximum speed is attained. [1]

[Total: 10]

8. [PJC/H2/Prelims 2018/P2/1]

- a. Define *work done*. [1]
- b. A trolley of mass 400 g is moving at a constant speed of 2.5 m s^{-1} to the right as shown in Fig. 8.1.

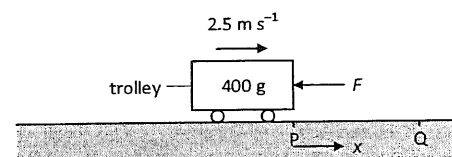


Fig. 8.1

A variable force F acts to the left on the trolley as it moves between points P and Q. The variation of F with displacement x from P is shown in Fig. 8.2. The trolley comes to rest at point Q.

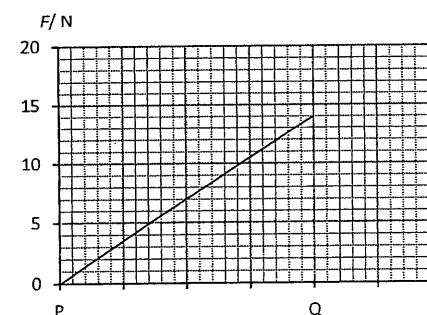


Fig. 8.2

- Calculate the distance PQ. [2]
- On Fig. 8.3, sketch, with appropriate values, the variation with x of work done on the trolley by F . [3]

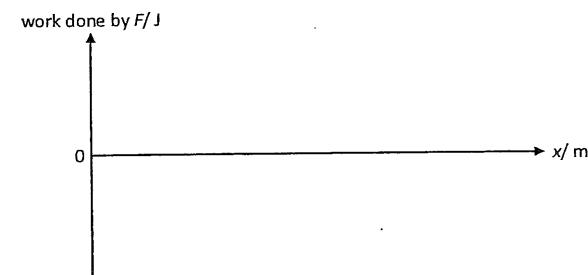


Fig. 8.3

- iii. In order to maintain a constant speed of 2.5 m s^{-1} , an electric motor attached to the trolley is switched on. On Fig. 8.4, sketch the variation with x of the power supplied by motor while the trolley moves from point P to Q. No numerical value is required. [1]

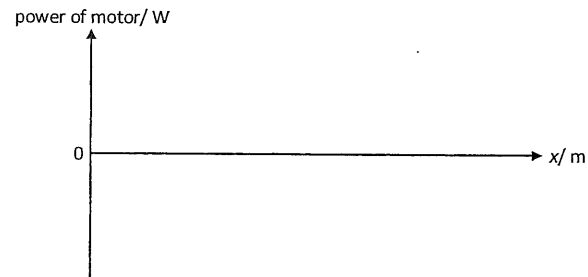


Fig. 8.4

[Total: 7]

9. [RVHS/H2/Prelims 2018/P3/1]

- a. A body of mass m is accelerated by a constant resultant force from rest to a speed v . By considering the work done on it, with reference to equations of motion, derive an expression for its kinetic energy E_k . [3]
- b. A car of mass 820 kg is travelling at constant speed along a horizontal road. The engine of the car provides 37 kW and a constant drag force of 1.2 kN acts on the car throughout its motion.
- Show that the work done in overcoming the drag force during a time of 11 s is 410 kJ . [1]
 - Calculate the maximum speed that the car can travel at if the engine provides half the original power. Show your working clearly. [2]

[Total: 6]

10. [VJC/H2/Prelims 2018/P2/2 Modified]

- a. Explain the various energy conversions that occur during each of the following events:
- A car moving on a rough horizontal road coming to a stop. [2]
 - A mass is attached to the bottom of a vertical spring. The mass is lowered slowly until it reaches an equilibrium position and remains stationary at that position. [3]
- b. Fig. 10.1 shows a skateboarder, of mass 55 kg , entering a ramp at point A with a speed of 6.5 m s^{-1} . He then leaves the ramp vertically at point B, 1.9 m above A. The distance travelled between A and B is 2.0 m and the average frictional force is 2.0 N . The centre of gravity of the skateboarder is 0.60 m above A.

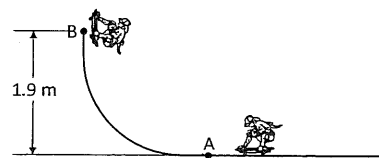


Fig. 10.1

- i. Determine the speed of the skateboarder as he exits point B. [3]
- ii. State and explain whether your answer to b.i. will be different if the ramp is replaced by the one shown in Fig. 10.2 below. The average frictional force by the ramp is still 2.0 N . The points A and B are in the same positions as in Fig. 10.1.

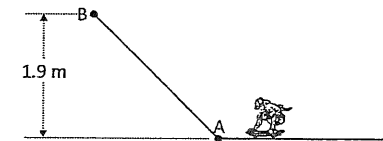


Fig. 10.2

[2]

[Total: 10]